PROJECT AND TEAM INFORMATION

Project Title

IntelliPaper: Intelligent Question Paper Generation System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation

The creation of a typical question paper is currently a manual process. The individual teachers write questions individually, hand them to the HOD or the concerned person for creating a question paper. Then the concerned HOD or Paper Setter has to hunt, scan, compile and arrange appropriate questions so that the entire paper could be set. On increasing the number of subjects, units, teachers, such work consume much time. There are many problems associated withmanual way of generating papers like usage of similar type of questions more than one time, not equal no of questions for one unit, then getting the paper with difficult questions only or with easy questions only, remembering which of the question is used last year, then for how many no of years that question has been used..etc. This proposed Intelligent Question Paper Generation system called IntelliPaper system will store all types of questions one central place known as question repository. Teachers will add individual question with certain properties, subjects, unit, topic, marks, difficulties level and bloom level, marks, difficulty level, bloom level etc. Then the HOD or concerned person will input into system with which type of questions we want, then the system will generate relevant question on the basis of the required parameters then at first the priority of those questions that are used less then question papers will contain fewer numbers of same questions. The primary goal of the project to reduce human work, to create question papers that are reliable, have fewer numbers of re-useable question. The project has following objectives: The main idea and objectives of this project was derived from the project report.

State of the Art / Current solution

At present, the papers are mostly developed from a question bank, Word files, spreadsheets, or questions sent from different teachers individually. The teachers write and submit the questions, then the HOD/Paper setter chooses questions manually and creates a paper by using their intuition and experience, considering criteria such as subject, mark, difficulty, unit, etc. Although digital question banks can facilitate searching for and storing questions but the choosing questions manually, arranging questions based on usage is time consuming and difficult to maintain properly a history of used questions, and specifically, ensuring questions used recently or too many times are least prioritized is challenging. The IntelliPaper system aims to automate question preparation to an extent by storing all questions and their details along with usage information in a single relational database and HOD just needs to input question paper requirements like number of questions, marks, unit, difficulty, etc., which the system retrieves and organizes based on usage criteria. The system will use the following starting priority formula:

Priority = 1 / (Times Used + 1)

So, a question never asked will have more priority than a question that's been asked multiple times. For example, for questions asked 0 times to questions asked 4 times, priorities will be 1 and 1/5 respectively.

In future iterations, the system can potentially be enhanced to also check: when last asked, all concepts and units are addressed, questions of varying difficulties exist, and random selection with control

Project Goals and Milestones

0.1 Project Goals

Project Objective

This system aims to provide a smart system which generates question papers automatically from a single master database where questions are stored. It selects different, appropriate questions to generate the question papers.

Specific Objectives

To create a single question repository which can hold questions from various teachers.

To store various attribute information of a question like subject, unit, topic, marks, difficulty level, bloom's level etc.

Assign distinct features for teachers, HOD/paper setters administrators according to the role. Use a priority based system for selecting questions based on how many times the questions were used. Don't use the same question in the last few question papers. Store the generated question papers and the selected questions for future use. Employ DBMS concepts like normalization, SQL, indexing, transactions and constraints for data management. Utilize OS concepts like concurrency, synchronization, critical section, locking and scheduling for secure management of concurrent users.

Initial Milestones

Milestone 1 — Requirement Analysis *Conduct requirement analysis to understand the manual creation process of the questions. Understand what the system should be performing in terms of functionality. Document the functional and non-functional requirements.* ***Milestone 2 — Database Design*** *define and identify relationship between them. Generate the ER model. Design the tables in the relational database so they are set up correctly. Define primary, foreign keys etc within the database.*

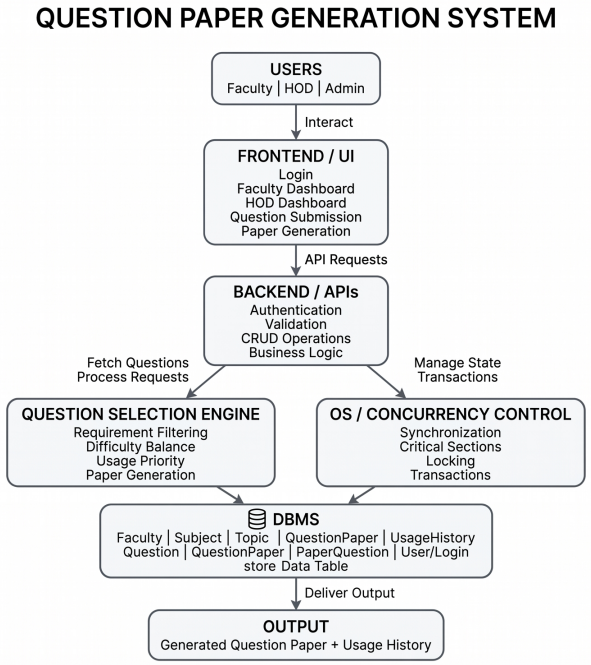
Milestone 3 — System Architecture *Work out the design for the front end, back end, the database and the questions that can be selected. Work out how to make all components talk to each other within the system.*

Milestone 4 — Core Algorithm *Core Algorithm Method of how tofind and filter suitable questions. Method for assigning points to questions based on usage history. Automatically generated test papers.*

Project Approach

A very basic ***frontend-backend-database architecture*** *is proposed for this system. The system is made up of modules and each module does a specific job.*
The Frontend is what users see and interact with.
Teachers can log in. Add questions to the question bank. They add information as they are typing in the question i.e. Subject, Unit, Marks difficulty level. The Head of department can set criteria for the question paper then look in the bank of questions in order to build the question paper. They are capable of reviewing, editing, and also submit the question paper. The Back end functions of the web system. It links the front-end of the web to the database. Performs functions that include adding, viewing, editing and deleting, saving also editing the question bank; login services; also generation of the paper. Data validity is used to verify if information given is correct before storing in the database. CRUD functions add, update, view delete from teacher and question table. The database stores all the information about teachers, subjects, units, questions, and the relation between them. Uses techniques such as the keys of the table, foreign keys, normalization, constraints, joins, indexes, transactions, SQL queries. For generation, when you pass specific values the questions matching that criteria are extracted and printed. The system forms the paper based on the selection.
Start Formula is given by: ***Priority = 1 / (Times Used + 1)***
This formula gives a higher priority to questions that have been used fewer times.
The ***Operating Systems component*** *This will contribute towards handling correctly the system when many concurrent users are active on the system. As an example several teachers might send or modify questions simultaneously or send many paper-generation requests simultaneously. Critical section concepts, synchronization, locking, transactions, elimination of race conditions, deadlocks prevention will be take into account.*
The project is composed in steps in order to be developed, the steps are to clarify the requirements, to plan the design, to develop various part of the system and finally to test all the test the system and verify the results. Suitable programming language(s), relational DBMS, frontend technology, IDE and Git/GitHub will be used according to final technology decisions.
The total workflow and relationship of the OS and the DBMS concepts will occur exactly as those described in the project documentation.

System Architecture (High Level Diagram)(2 pts)



0.1.1 Architecture Explanation

The system will be comprised of 4 main layer: ***Frontend***, ***Backend***, ***Intelligent Selection Engine***, ***Database***. *It provides a method that allow Teachers and HODs to work with the system. Users can log in the system and working based on Teachers requirements and HODs' requirement. The backend system fulfills these requirements. As the backend system knows when to do and when not to do it. The system extracts questions according to the HODs requirement to form the question paper.*
Stores the details of the questions:
History of used question. Maintain all the persons simultaneously ensuring that the Database cannot be corrupted with the lose of any data.

Project Outcome / Deliverables

The expected outcome of the project is a functional ***Intelligent Question Paper Generation System*** *that automates major parts of the question-paper preparation process.*
Major Components

1. ***Question Repository*** *A place where teachers can save and manage questions .*
2. ***Relational Database*** *Stores information about teachers, subjects, topics, questions, question papers, and question usage history.*
3. ***Faculty Module*** *Allows teachers to add, edit, and view their questions.*
4. ***HOD/Paper Setter Module*** *Allows the HOD to enter the requirements for the exam and create the question paper.*
5. ***Question Filtering Module*** *This module searches for a question based on subjects, units, marks, difficulty, status etc.*
6. ***Intelligent Priority Algorithm****Prevents repeated selection of questions and prioritizes those questions which were once in use and not again after sometime.*
7. ***Question Paper Generation Module****for producing the final paper.*
8. ***Usage History Module****for maintaining records of previously used questions.*
9. ***Concurrency and Transaction Handling****to maintain data integrity during simultaneous operations.*
10. ***Testing and Evaluation Report****containing test cases and system results.*

The final system should demonstrate practical integration of DBMS and Operating Systems concepts while reducing manual effort and improving the reliability and diversity of generated question papers.

Assumptions

A few implicit assumptions are made in the designing of the system. Faculty Members can be assumed to access the system with their valid user IDs and provide all valid questions with appropriate meta data as and when desired by them. HOD or competent Paper setter is expected to specify the paper details viz subject, number of units to be covered, total marks allotted, the desired number of questions and the question paper would contain a distributed number of questions based on the level of difficulty. We assume that all the questions residing in the data base have been thoroughly verified by some appropriate authority so as to incorporate them into the paper generation.
We have considered that a relational DBMS is to be provided, and the appropriate persons are granted sufficient permissions to access and modify it.
As many different types of users would be accessing the system at the same time, database transactions and locking would exist for this system.

References

1. *Silberschatz, A., Galvin, P. B., & Gagne, G., Operating System Concepts.*
2. *Korth, H. F., Silberschatz, A., & Sudarshan, S., Database System Concepts.*
3. *MySQL Documentation — Official documentation for relational database implementation.*
4. *PostgreSQL Documentation — Official documentation, if PostgreSQL is selected.*
5. *Documentation for the selected frontend/backend technologies.*
6. *References related to* ***Bloom's Taxonomy*** *for question classification.*
7. *Research papers related to automated question-paper generation and question-selection systems.*